

PAPER_524 — Plasma Orb Emergence Threshold: Probabilistic Model and Orion Nebula Proplyd Calibration

Unified Quantum Field Framework (UQFF)

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§1 — Novel Physics Claim

Plasma orbs are **emergent structures** in the lower 1/3 stable end of the Universal Spectrum, serving as direct precursors to cosmic quantum eggs. Emergence is governed by a probabilistic threshold condition:

$$US_{\{\text{orb}\}} > \langle US_{\{\text{orb}\}} \rangle + \sigma(US_{\{\text{orb}\}}) \cdot \text{Prob}_{\{\text{order}\}}$$

When cumulative spectral energy $US_{\{\text{orb}\}}$ exceeds this threshold, a plasma orb separates from the spectral continuum and begins its transition toward a quantum egg.

UQFF acts as a **post-hoc encompassment** (not causation): the observed proplyd properties ($\$r$, $\$M$, $\$ \dot{M}$, $\$v$) are encompassed within the UQFF Buoyancy Gradient framework to within 10% residual.

§2 — Master Equations

Emergence Threshold

$$\text{Emergence if: } \quad \text{US}_{\text{orb}} > \mu_{\text{US}} + \sigma_{\text{US}} \cdot \text{Prob}_{\text{order}}$$

Buoyancy Gradient (Spectral Form)

$$\text{Buoy}_{\text{grad}} = \frac{\rho_{\text{UA}} \cdot V_{\text{displaced}} \cdot (F_{\text{inert}} + \text{Resonance}_{\text{harm}})}{1 + \Delta_{\text{dil}}}$$

Vacuum Density Series Anchor for ρ_{UA} (PAPER_429)

$$\rho_{\text{UA}}^{\text{anchor}} = \sum_{n=1}^{\infty} \frac{[\text{SSq}]^n}{n^{26}} = \text{Li}_{26}([\text{SSq}]) \approx 0.570$$

This anchors the stable lower-1/3 boundary: ρ_{UA} in the Buoyancy Gradient is bounded from below by the Vacuum Density Series convergence value.

Emergence Fraction

$$f_{\text{emerge}} = \frac{N_{\text{emerged}}}{N_{\text{total}}}$$

§3 — Numerical Results

Calibrated against the Orion Nebula proplyd population (Hubble / MUSE):

Quantity	Value
Emergence threshold	$3.62 \times 10^{16} + 4.10 \times 10^{16} \times 10^{-4}$
Emerged fraction f_{emerge}	18.32%
Mean proplyd size	375.87 AU (obs: 250–500 AU ✓)
Mean mass	0.63 $M_{\odot}$
Mean mass-loss rate	$4.67 \times 10^{-6} M_{\odot}/\text{yr}$
Mean velocity	9.76 km/s
$\text{Li}_{26}([\text{SSq}])$ anchor	≈ 0.570 (50-term sum)
Residual budget	$< 10\%$

§4 — Standard-Model Comparison

Classical proplyd models (Johnstone et al. 1998, Störzer & Hollenbach 1999) attribute proplyd structure to external UV photoionization:

Classical UV Model	UQFF Encompassment
External photoionization	Spectral threshold $US_{\text{orb}} > \theta$
Mass-loss from UV flux	$Buoy_{\text{grad}}$ drives $\dot{M}$
No frequency structure	Lower 1/3 stable spectral regime
Single-epoch rates	$Prob_{\text{order}}(t_{\text{neg}})$ evolution

UQFF does **not** replace the UV model — it provides a spectral framework within which the observed 18.32% emergence fraction and proplyd properties are naturally reproduced.

§5 — Testable Predictions

1. **18% emergence universality:** Surveys of proplyd-hosting HII regions other than Orion (e.g., Carina Nebula, W49) should find emergence fractions consistent with $f_{\text{emerge}} \approx 0.18$, if the $[SSq]$ anchor is universal.
2. **Threshold scaling with $[SSq]$:** Systems with different calibrated $[SSq]$ values should show emergence fractions $f \propto Li_{26}([SSq])$.
3. **Buoy_grad mass-loss correlation:** The quantity $Buoy_{\text{grad}} / (1 + \Delta_{\text{dil}})$ should correlate linearly with observed proplyd mass-loss rates across the Orion sample.

Cross-reference: PAPER_429 (Vacuum Density Series $Li_{26}([SSq]) \approx 0.570$); PAPER_521 (US Spectral Divisions 1/3 boundary); PAPER_523 (Quantum Egg Sim)